McGill initiative in Computational Medicine

McGill | CDSI Computational and Data Systems Institute | ISCD Institut en systèmes computationnels et de données

D2R DNA TO RNA | HeDS

Canadian Centre for Computational Genomics

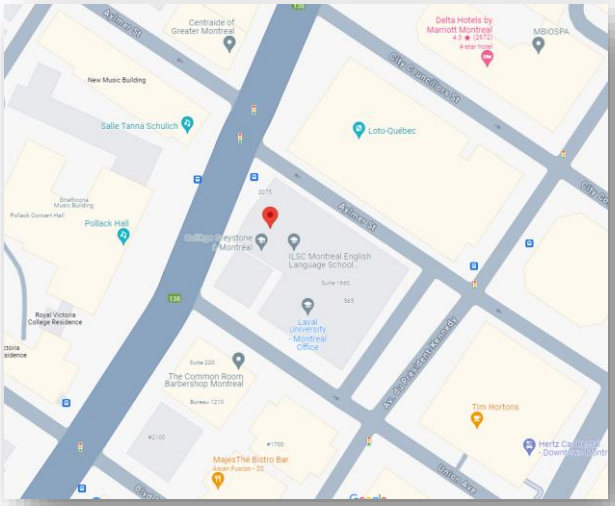
CODE TO MANUSCRIPT BOOTCAMP · DAY 4

Building figures with Inkscape

Instructor: Ariel Madrigal
Facilitator: Senthilkumar Kailasam
October 29, 2026

McGill initiative in Computational Medicine

QLS-MiCM mission statement: deliver quality workshops designed to help biomedical researchers develop the skills they need to succeed.



Location: 550 Sherbrooke Street, Montreal, Quebec

https://www.mcgill.ca/micm

Outline:

1. Introduction to scientific figure design and Inkscape
2. Importing and editing plots
3. Building a multi-panel figure
4. Exporting and closing remarks

This is an interactive workshop :)

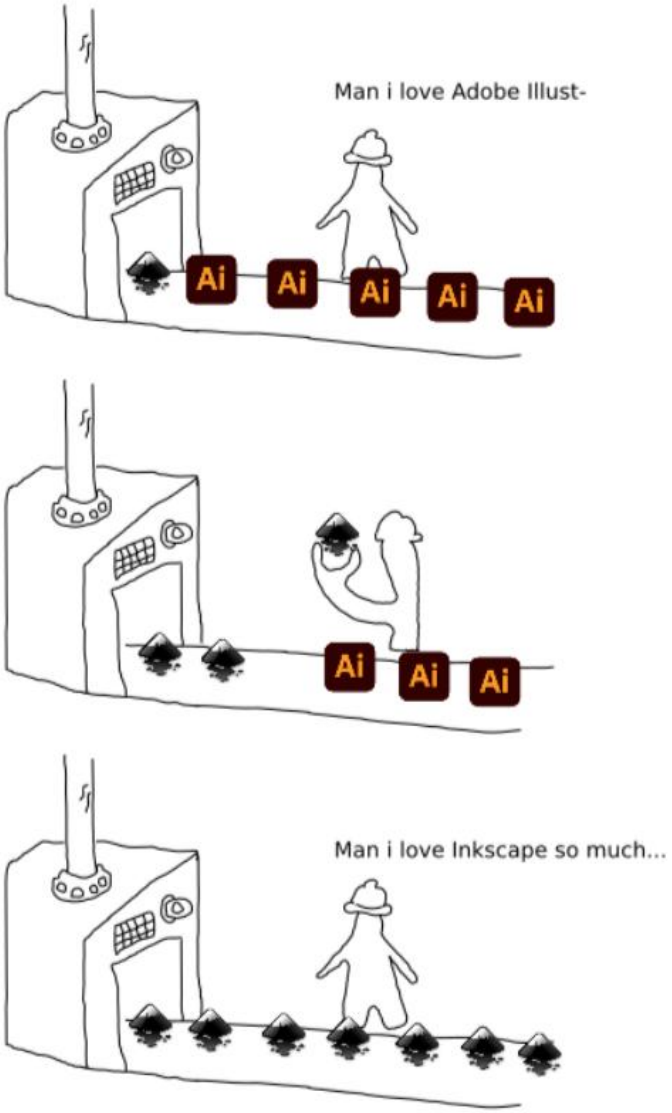
Feel free to interrupt or raise your hand to ask questions

The objective for today

To use Inkscape to transform a collection of independent plots into a polished, publication-ready figure

Part 1: Introduction to scientific figure design and Inkscape

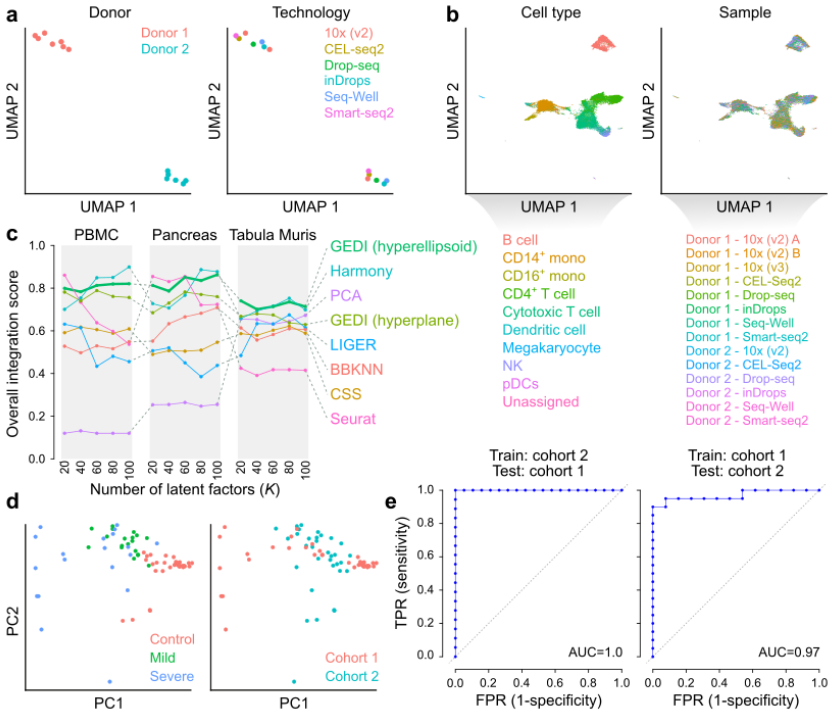
Why use Inkscape for scientific figures?



- It's free!
- Open source
- Cross-platform (Mac, Windows, GNU/Linux)
- Supports vector graphics
- Enables creation of high-resolution figures

What can you do with Inkscape?

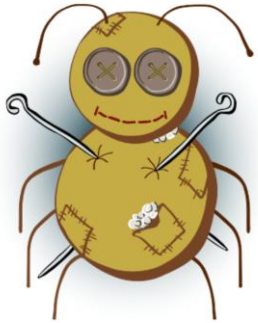
Scientific figures

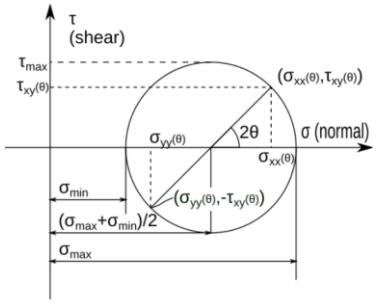


But also, much more!

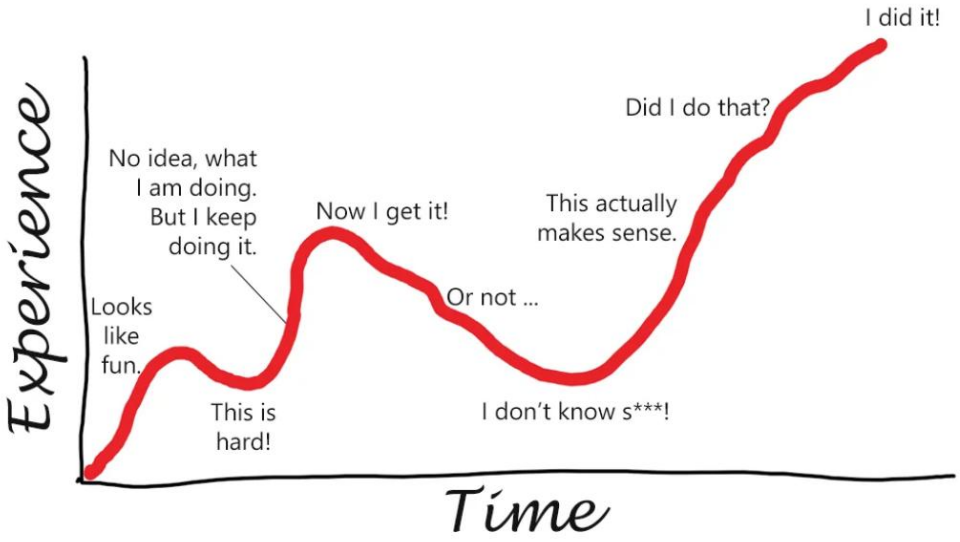








Inkscape can be hard…



But it's worth it

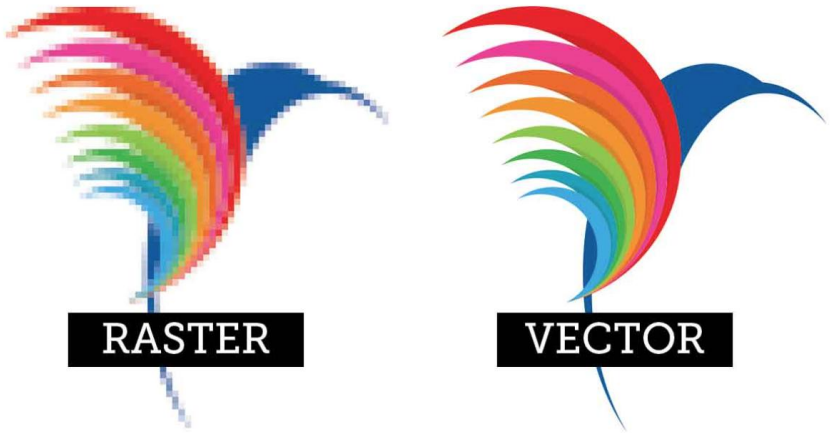
Vector vs raster graphics

Raster graphics

- Made of **pixels** arranged in a grid
- Resolution-dependent
- Become blurry or pixelated when enlarged
- Best for photographs and complex images
- Larger image dimensions are needed for high-quality printing

Vector graphics

- Made of **mathematical shapes and paths**
- Resolution-independent
- Can be scaled to any size without losing quality
- Individual elements remain editable
- Ideal for plots, diagrams, illustrations, and figure assembly

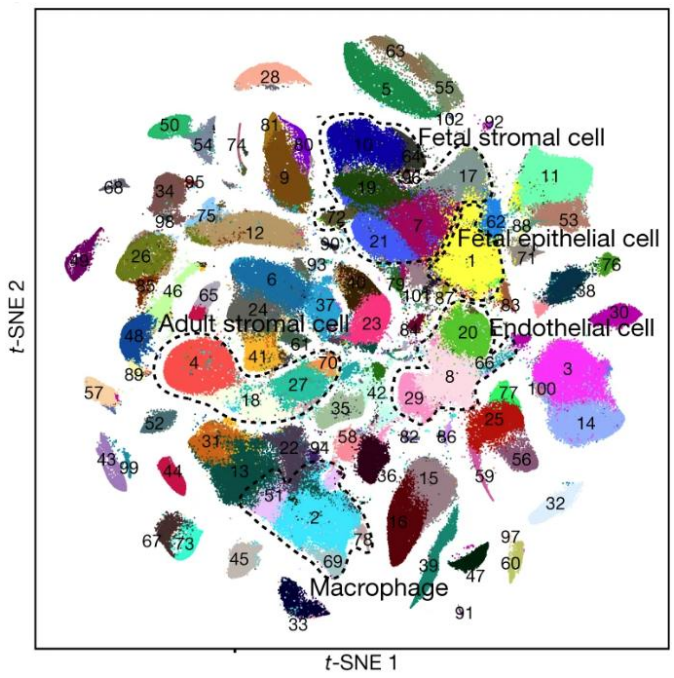


When to use vector or raster images?

Raster graphics

Complex technical images:

- Microscopy images
- Histology slides
- Plots with hundreds of thousands of points



Vector graphics

When possible, use vector-based graphics:

- Most of the computer-generated plots
- Diagrams and schematics
- Text, labels, arrows and annotations

Either:

- Use rasterization options while exporting your figures (e.g. ggrastr in R)
- Export rasterized versions of the plot without labels

Common file formats

Raster graphics

- PNG (.png)
- TIFF (.tif, .tiff)
- JPEG (.jpg, .jpeg)

Vector graphics

- SVG (.svg)
- PDF (.pdf)
- EPS (.eps)

Note:
If exporting as .svg in R, use the svglite package:
svglite::svglite(file, width = width, height = height, fix_text_size = FALSE)

Best practices for publication-quality figures

Use vector formats whenever possible to preserve quality and editability

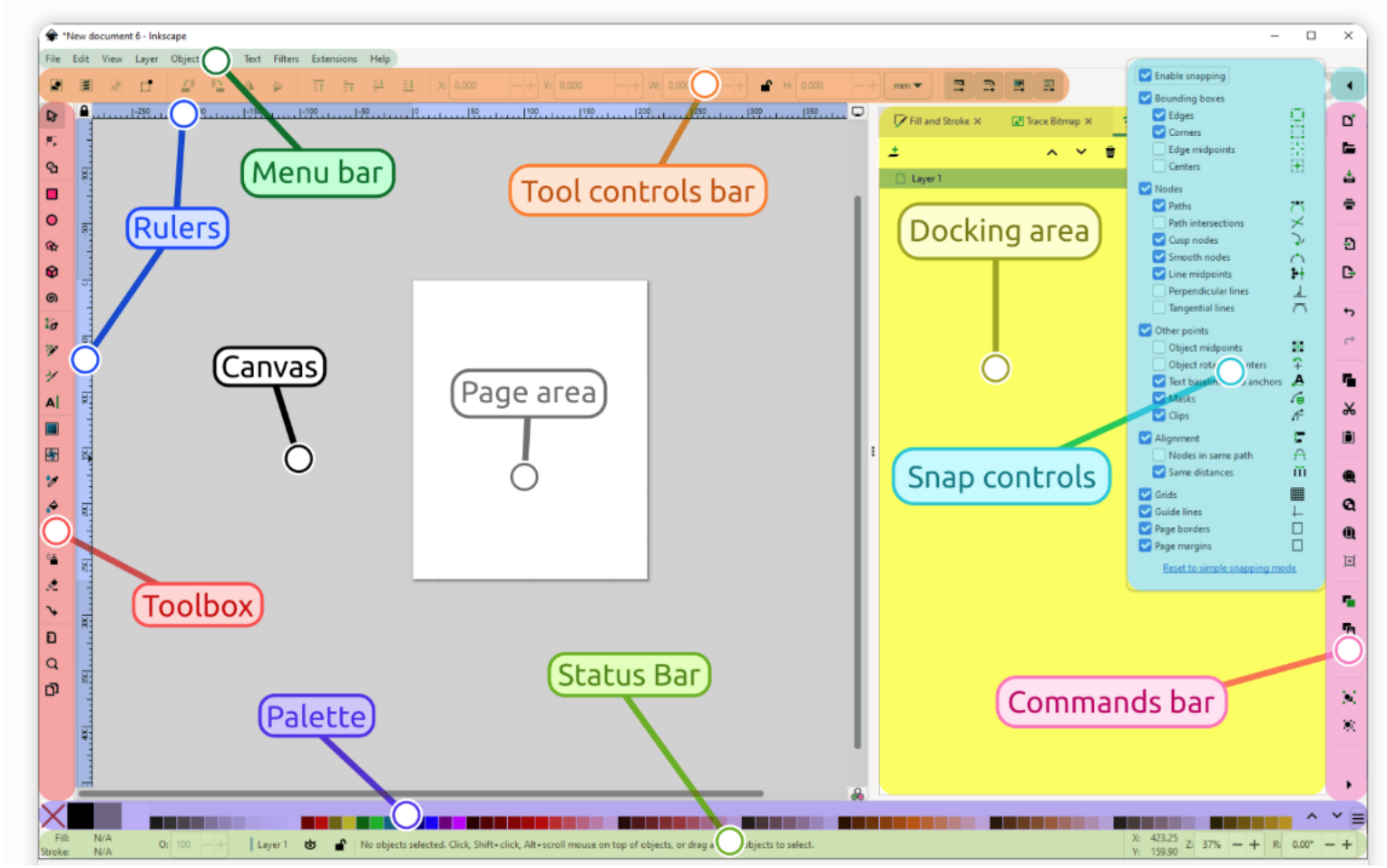
Ensure text, labels, and legends remain readable

Align panels and use consistent spacing to create a balanced layout

Use color thoughtfully and consider accessibility (e.g., colorblind-friendly palettes)

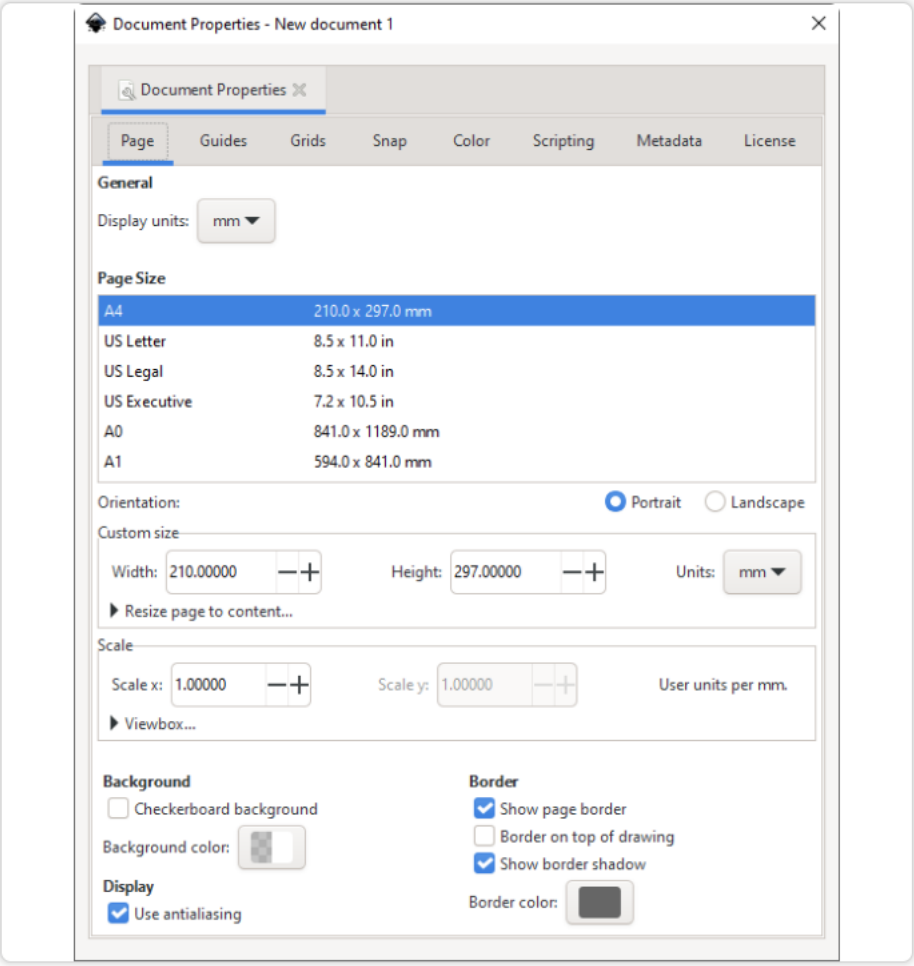
Export figures using journal-appropriate formats and dimensions

Overview of the Inkscape interface



Manage the document properties

Let’s go to File -> Document Properties

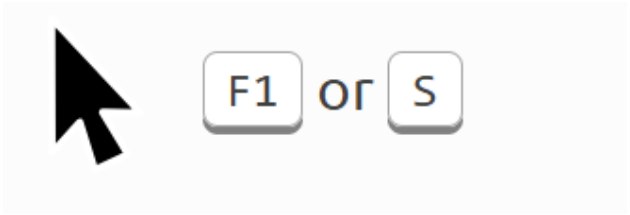


1mm == 3.78 px

Part 2: Importing and editing plots

Let’s familiarize with some tools

Selector tool



Zoom tool

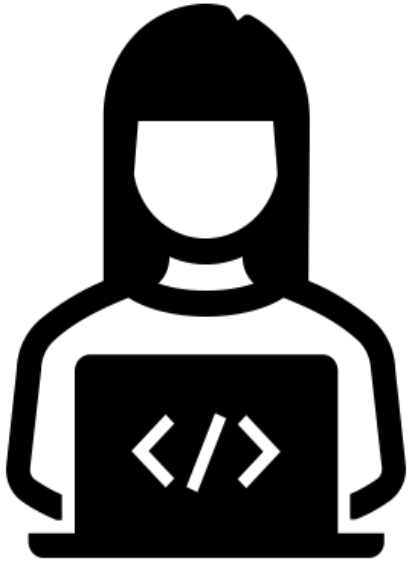


F3 or Z

Text tool



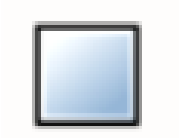
F8 or T



Inkscape shortcuts
https://inkscape.org/doc/keys.html

Let’s create some basic shapes

Make rectangles



R

Make circles, ellipses and arcs

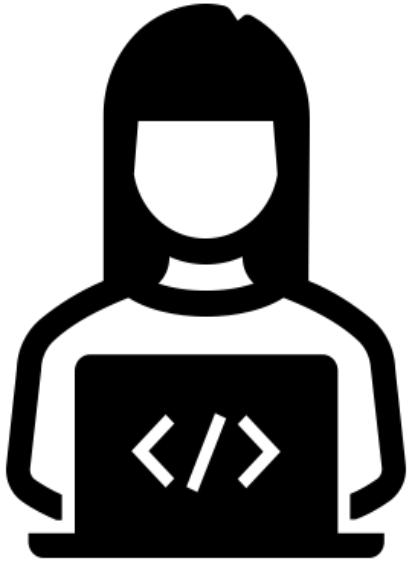


E

Make polygons, stars



*

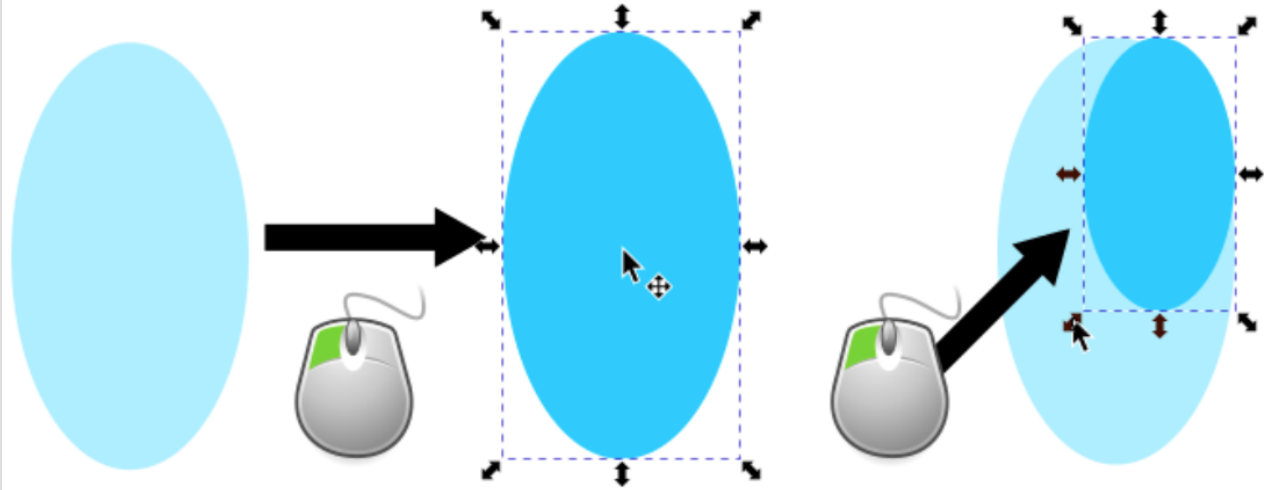


Inkscape shortcuts
https://inkscape.org/doc/keys.html

Moving, scaling, skewing and rotating objects

The Selector Tool

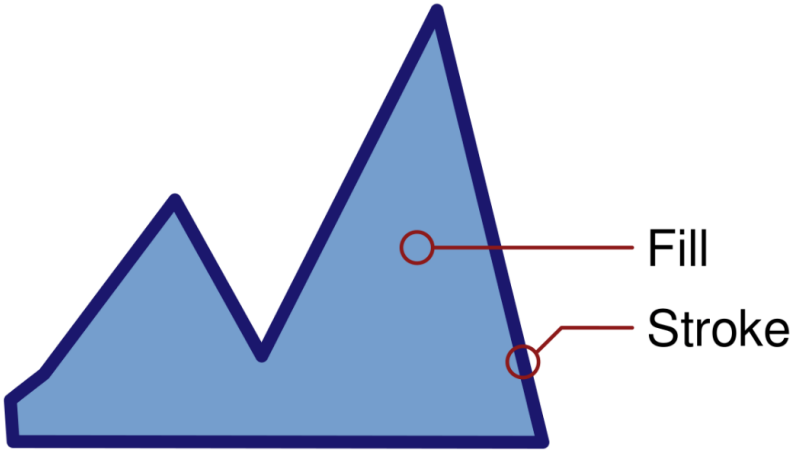
F1 or S



Drag to move the selection. Transformations (such as moving, scaling, rotating) are easy thanks to the two-way arrows.

Colors

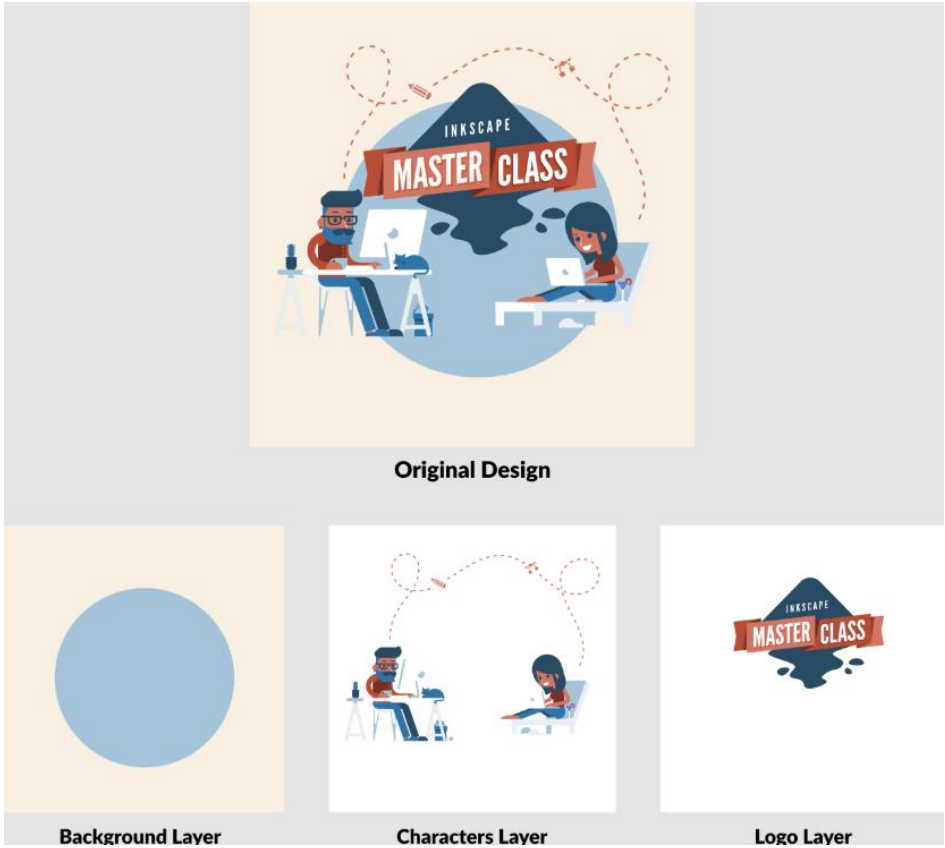
For a vector object, a separate style can be defined for its outline (or stroke) and its main color (or fill). This can be a plain color (e.g. dark red), a pattern (e.g. stripes or flowers), or a gradient (e.g. a smooth transition from green to blue).





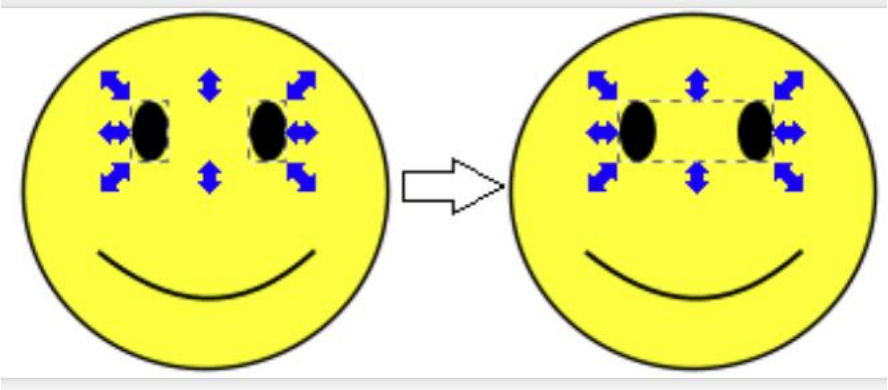
Understanding layers

Layers allow you to organize elements of your figures into sections

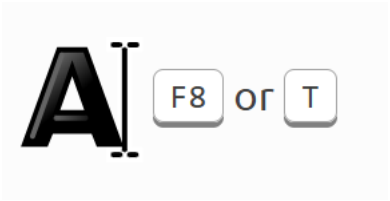


Group objects

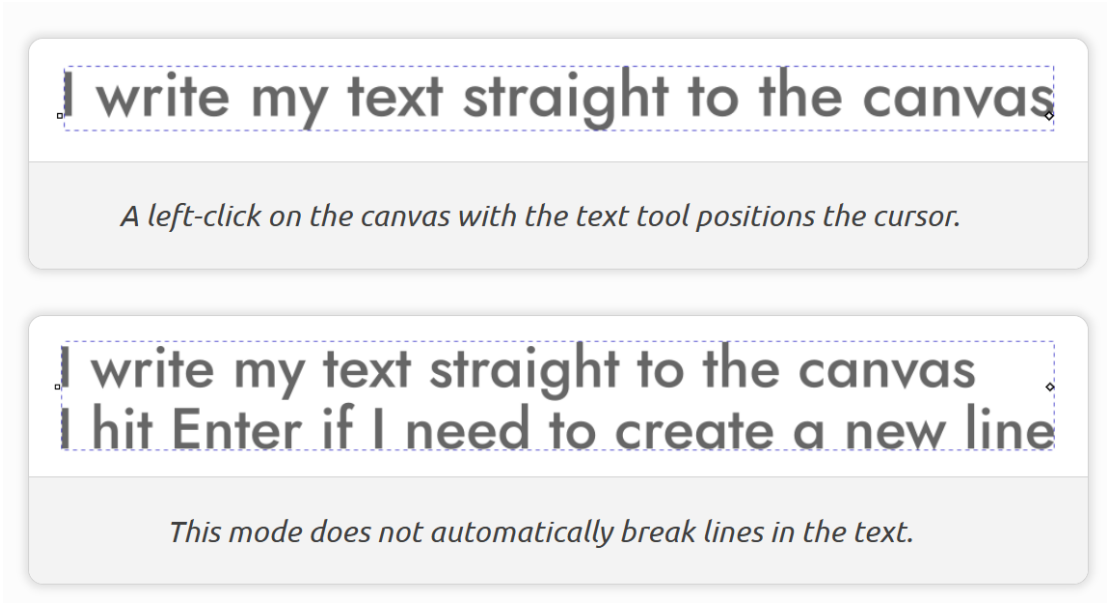
Grouping allows you to work with multiple objects as a single object: any changes you make will equally affect all members of the group, and the entire group can be copied, duplicated or cloned as a unit



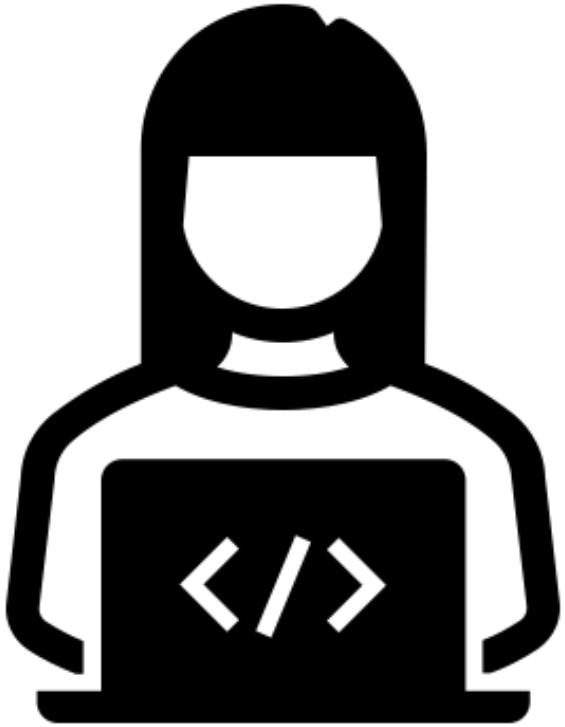
Managing fonts and text elements



Two options of how to create text:



Importing images into Inkscape



Go to Exercises/data:

Import the same figure in different formats:

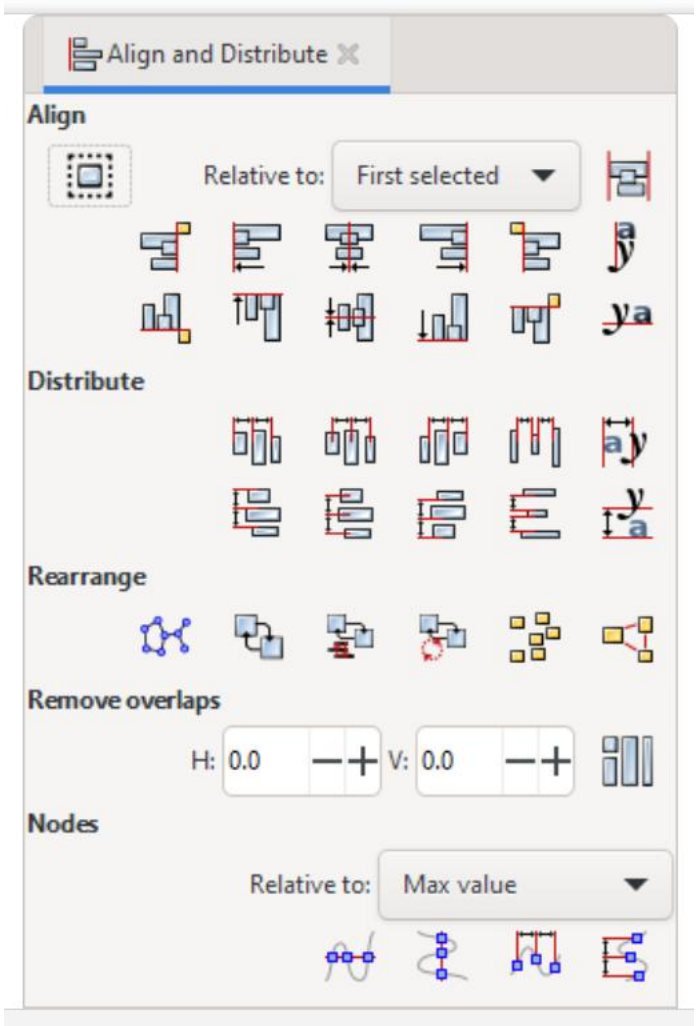
- PCA (.png, .pdf, .svg)
- PCA rasterized (.pdf, .svg)

Part 3: Building a multi-panel figure

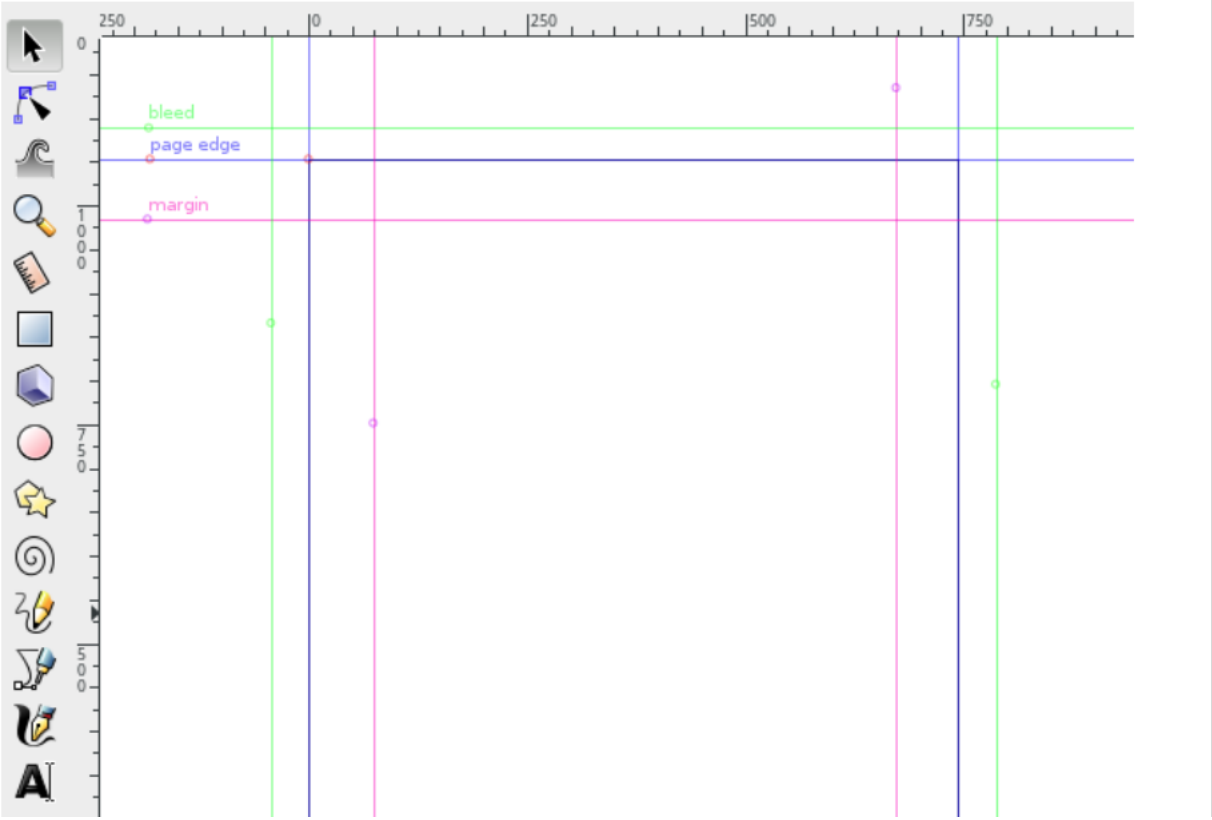
Alignment and distribution tools

Align and Distribute

Shift + Ctrl + A



Guides



Copying a color

There are several ways to apply the same style to multiple objects:

- **Selecting multiple objects**: select all the objects whose color you want to change (hold `Shift` and click, or drag a selection box around them), then open the `Fill and Stroke` dialog, select a color. It will be applied to all selected objects automatically.
- **Use the Dropper tool**: select an object, then activate the Dropper tool by clicking on its icon 🖉, then click on the canvas, on an area where it has the color that you want to apply. The selected object's color will change accordingly.
- **Copy the complete style of an object to apply it to another object:** select the object, copy it to the clipboard with `Ctrl` + `C`, then select the second object and paste only the style with `Shift` + `Ctrl` + `V`. This not only copies the fill and stroke color, but also the stroke width and style, a text's font style and any patterns or gradients.

Clipping and masking

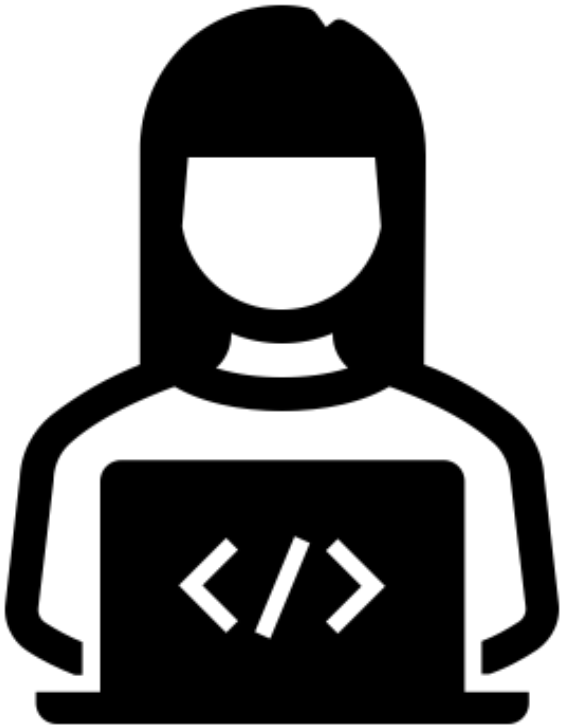
Put the object that will serve as clip (or mask) above the object that you want to clip (or mask)



Clipping

Masking

Let's build a multi-panel figure



Import your own PDF figures generated in R or Python, either from previous workshops or from your own projects. For those who do not have their own figures:

Go to Exercises/data:

Import the following figures:

- PCA_rasterised.svg
- Plot_expression.pdf
- Heatmap_geneExpression.pdf
- EnhancedVolcano_raster.svg

Assemble a figure

Part 4: Exporting and closing remarks

Saving and exporting

Always save your current work as .svg (Inkscape .svg)

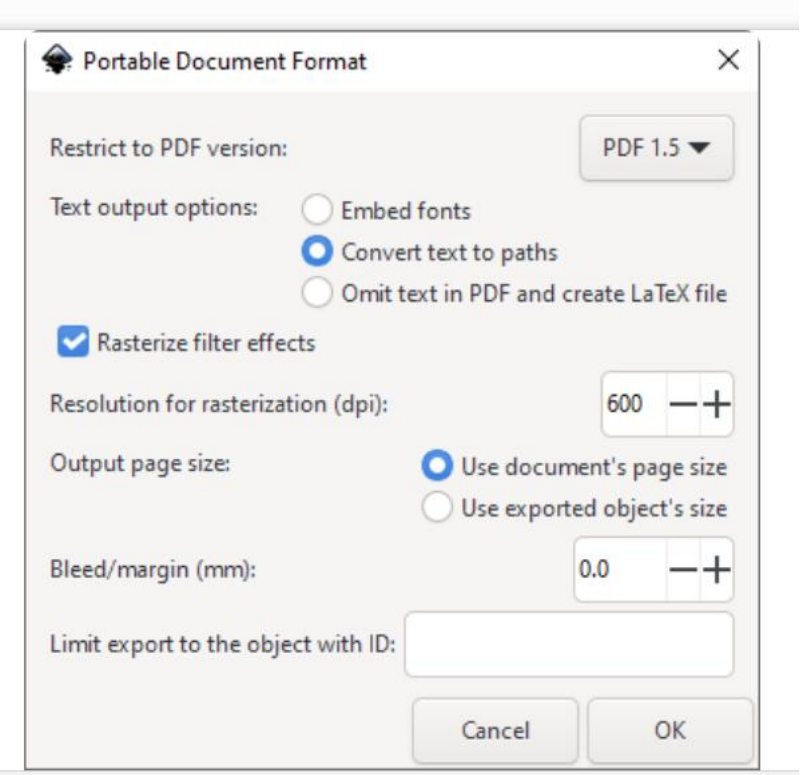
For exporting, you have several options:

- For high-quality figures (Save a copy to PDF)

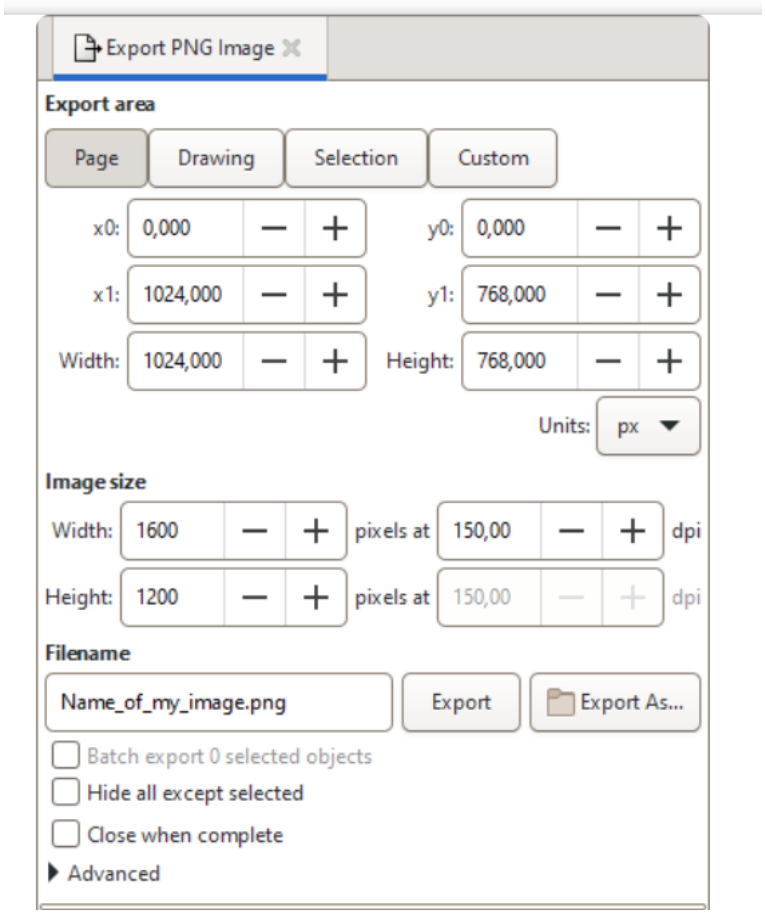
- For raster graphics file format, export to one of the following formats:
    - PNG, JPEG, TIFF

Saving a PDF

Never save as PDF only, but always keep an SVG file, because the PDF file format supports a different set of features from the SVG file format and you may lose data in the process



Exporting a PNG file



Some useful resources:

Inkscape Beginner's Guide

Inkscape manual and books

NIH BIOART source

Please share with us your feedback!



Thanks for your attention!

McGill University

Keep an eye for the workshops offered by the MiCM!

info-micm@mcgill.ca
https://www.mcgill.ca/micm/

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Canadian Centre for Computational Genomics

D2R DNA TO RNA | HeDS

References

- Inkscape manual. https://inkscape-manuals.readthedocs.io/en/latest/index.html
- Zabala A. *Designing Effective Scientific Figures: Practical Inkscape Tutorial*. Bioinformatics Core Shared Training. https://bioinformatics-core-shared-training.github.io/effective-figure-design/DesigningEffectiveScientificFigures_Practical_INKSCAPE_Zabala_v00.pdf